

Since Column 3 must have a pivot, From condn. 2 of the matrix to be in Echelon form, Row 3 Column 3 has the pivot, i.e. Row 3 Column 3 = $\square$.
By condn. 3 of the matrix to be in echelon form, Row 4 Column 3 = 0.

$\therefore$ Final echelon for the matrix:

$$\begin{bmatrix} \square & * & * \\ 0 & \square & * \\ 0 & 0 & \square \\ 0 & 0 & 0 \end{bmatrix}$$

Suppose, Row 1's ~~not~~ pivot is not in Column 1. $\therefore$ At least one of Row 2 to Row 4 has its pivot in Column 1. This violates condn. 2 of the matrix to be in echelon form. $\therefore$ Row 1's pivot must be in Column 1. $\square$

27) How many pivot columns must a 7×5 matrix have if its columns are linearly independent? Why?

Solution: Let A be a 7×5 matrix whose columns are linearly independent.

- $\therefore$ The matrix equation $Ax=0$ has only the unique trivial solution.
- By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.
- By row reduction algorithm and Theorem 1, the augmented matrix $[A \ 0]$ can be converted to a ~~equivalent~~ rref which is unique.
- By Theorem 2, the matrix A has no free variables.
- $\therefore$ All columns of matrix A are pivot columns $\square$

28) How many pivot columns must a 5×7 matrix have if its columns span $\mathbb{R}^5$? Why?

Solution: Let A be an arbitrary 5×7 matrix. Suppose, the columns of A span $\mathbb{R}^5$. By Theorem 4, all the 5 rows of A have pivot positions.

By defn, a pivot position corresponds to the leading 1 of a row in rref form. From the row reduction algorithm and Theorem 1, the matrix A can be converted to a unique rref. From condn 2 of rref form, the pivot position of each row must be in a column to the right of the columns of the pivot positions of all rows above it. $\therefore$ It's easy to see that there must be 5 pivot columns. $\square$